

AHMED RAYHAN

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Research Activities

Ahmed R., Z.I. Bushra **Enhanced Blood Glucose Detection Using SPR-Based Biosensor with CdS/Ag/CdS/BP₂ Nanomaterials**

IOP Physica Scripta-2025, Impact Factor : 2.6, Quartile: Q2.

DOI: 10.1088/1402-4896/addfb3

In this work, we present a cutting-edge biosensor leveraging surface plasmon resonance (SPR) technology integrated with BaF₂/CdS/Ag/CdS/BP nanomaterials for superior blood glucose detection. The performance was validated through simulations using MATLAB and COMSOL Multiphysics, showcasing minimal error and excellent reproducibility.

Ahmed R., Emon W., N. Zaman. **Detection of Tuberculosis in Blood Samples Using One-dimensional Photonic Crystal**

Optical and Quantum Electronics - 2025, Impact Factor : 4, Quartile: Q2.

DOI: <https://doi.org/10.1007/s11082-025-08347-1>.

This paper presents a 1D binary photonic crystal biosensor utilizing MATLAB simulation for detecting Mycobacterium tuberculosis (TB) in blood samples, utilizing refractive index variations to achieve high sensitivity and specificity. The sensor's design, optimized using the transfer matrix method, making it a promising tool for early TB diagnosis.

Ahmed R., Smita C., B. Barua et al. **Performance Analysis of Free Space Optical Communication in Urban Rain Environment**

IEEE ECCE-2025 Conference Proceedings, Scopus indexed.

DOI: 10.1109/ECCE64574.2025.11013976

This paper investigates the feasibility of Free Space Optical (FSO) communication systems in urban Dhaka, Bangladesh, focusing on the impact of rain on signal performance. It concludes that FSO communication is viable in urban environments with proper mitigation strategies for rain attenuation.

Education

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Science in Electrical and Electronic Engineering

CGPA 3.559/4.00

Dhaka, Bangladesh

January 2025

HIGHER SECONDARY SCHOOL CERTIFICATE

Birshrestha Noor Mohammad Public College

GPA 5.0/5.0

Dhaka, Bangladesh

July 2019

Projects

- DC Motor **Speed and Direction** Control using Mosfets
- Implemented **Schematic and Layout** using virtuoso to implement Conventional Fat tree encoder.
- Home Security System using FSM and executed **Full Chip Design** with cadence Innovus & Genus.
- Automatic **Voltage Stabilizer**: Built device to regulate AC power output.
- LED Thermometer: Constructed a digital thermometer using **analog ICs**.
- **RFID** Locking System: Designed access control system.
- Gas Leakage Detection: Created **IoT** based system for gas leak detection.

Experience

Bangladesh University of Business and Technology

Lecturer

- Basic Electrical Circuits Theory and Lab, Power Electronics Lab

Dhaka, Bangladesh

June 2025- Present

DHAKA POWER DISTRIBUTION COMPANY

Attachment Trainee Intern

Dhaka, Bangladesh

October 2023 – November 2023

- Observed DPDC's organizational structure, including SCADA system, call center, and voltage switching.
- Participated in field visits to understand grid operations and efficient power distribution.
- Gained practical insights into Country's power distribution network.

Leadership and Activities

AUST Career Development Club

Dhaka, Bangladesh

Executive Director

April 2024 – Present

- Manage all IT department functions: strategic planning, project management, and team supervision.
- Lead major event planning and execution, maximizing attendee engagement.

Huawei Seeds For the Future

April 2024-July 2024

- Developed leadership, soft skills, and a winning IoT-based SDG project while networking with industry professionals.

Honors & Awards

- Achieved top 5% academic ranking in Spring '20, Fall '20, and Fall '23 semesters.
- Champion of IEEE's **Research Revelry: EV batteries**
Issued by IEEE AUST STUDENT BRANCH · May 2024
- Champion of IEEE's **Poster Pavilion – A Showcase of Innovation**
Issued by IEEE AUST STUDENT BRANCH · Jun 2023
- 7th Place in **Research Article Writing Competition**
Issued by IEEE AUST STUDENT BRANCH · Aug 2023

Skills & Interests

Technical	: MATLAB, COMSOL Multiphysics, CST Studio, Machine Learning, Python, C++, Virtuoso, Innovus, Genus, Arduino.
Soft-skills	: Problem Solving, Time Management, Leadership, Adaptability.
Interests	: Researching cutting-edge technology, reading fictions & travelling.

References

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